APS: four lead implementations and an instruction audit

This continuation implements the four proposals from the lead report, exposes them under the new `ic.aps_v3` identity, and extends the separate ordinary-residue reader. Development observations have producer standing, **unverified**; they are not benchmark data or a signed PCC verdict [pcc:unverified].

Deliverables: editable diagram, report PDF, proof map, cost contract, and design draft.

Requested step	Delivered component	Still required
1. Measurement readiness	Explicit lead controls; literal cells/history; separate arithmetic/full-set reader; optional method-plus-scalar Ir reconstruction	Actual profile/metric admission, complete GAE pricing, pilot, fixed analysis and frozen main protocol
2. Approved source checkpoint	Approval preserved; source-only GitHub gate and pinned selection support prepared	Fresh fenced admission, actual existing secret binding and successful standard gate; preparing code is not an upload
3. Baseline measurements	Original comparisons and four-size workload requirement retained in `PROTOCOL.md`	Successful prerequisites and actual EKS pilot/main execution; no performance numbers supplied
4. Implement and compare leads	Native trace pool, two-variable line chart, unequal profiles/arbitrary guesses, retained-base full/frontier controls and separate reader	Admitted measured comparisons; development tests do not establish improvement

Implementation

- **Trace reuse:** each target starts with an empty bounded pool. Existing native F4 record/replay supplies the candidate basis; numerical input reductions and every Buchberger S-pair must reduce to zero. An invalid, mismatched or resource-capped candidate falls back to ordinary F4. Learning, replay, checking, fallback and retained trace words remain visible.
- **Line chart:** for $y = tx + c$, use $U = t^2$, $V = c/t$ when $t \neq 0$. The divisor coefficients become $e_1 = U$, $e_2 = a - 2UV$, $e_3 = UV^2 - b$. The chart uses two unknowns; a separate one-variable horizontal cell is retained unless the existing propagated-orbit label certificate allows omission. Ordinate-zero and zero-denominator cases are refused. The existing unpruned line formulation also uses two variables. The certified pruned formulation uses a third inverse variable, which this chart removes; the protocol distinguishes these two comparisons.
- **Profiles and elimination:** original partition sizes are explicit, bounded and may differ. Guess any original coordinate before F4 and restore it before relation recovery. No unknown logarithm is supplied to a solver.
- **Growth:** preserve points and labels, append conditional new draws, and keep partial sampling failures. `retained_full` and `frontier` share the same nested base law. The latter skips old tuples only after a complete, non-timeout, non-early-stop traversal without an informative witness.

The implementations are bounded development backends: prime fields up to 65,521, arity three/four, original partition sizes one through four, degree bound at most 24 and explicit finite attempt/draw/time budgets. Cubic orbit expansion uses its existing independent certificate. No cap is lifted here. The existing six encodings and frozen v1/v2 records retain their behavior.

The independent reader validates actual producer records using ordinary residue affine arithmetic. It checks the partition/guess/chart cell geometry, literal digests, retained labels/draw totals, trace counter continuity and complete root sets through finite point combinations. A complete frontier's expected set is the new product's solutions minus the certified old product's solutions. A partial report only supports subset checking. This is separate software, not signed execution or raw-profile authenticity [pcc:unverified].

Cost and formal boundaries

ACCOUNTING.md defines optional inclusive Ir evidence and the excluded work. All operation-cost/execution flags remain honest: native GAE totals are incomplete; an unkeyed receipt is not execution proof. The new frame reader does not turn test duration into timing evidence.

PROOFS.md supplies the mathematical argument and proof map. The central frontier cover/disjointness statement has a core-only Lean formalization. Chart lifting into actual finite-field curve relations and the implementation-to-proof refinement remain open. The previous native identity records are reused at their original scope, never treated as PCC solve certificates or full chart-completeness proofs [pcc:unverified].

PROTOCOL.md is a design continuation, **not a freeze**. It keeps the original baseline comparisons, four held-out size classes, same-input references, controls, failures and complete cold-cost obligations. The source checkpoint is approved; local authentication is not bound and the repository secret path has not yet produced a gate outcome. No benchmark, runtime pilot, profiler run or EKS experiment was launched.

Evidence and continuation

SOURCE_CHECKPOINT.md describes a checkpoint-only GitHub path using the repository's existing secret reference. It requires a fresh T10 fence, exact source selection and a bounded upload window. Preparing that path is not a source upload, a gate pass or a benchmark admission.

`validation/early-boundaries.json` preserves the earlier failed and superseded development observations under native claim `PCC-20261106-e8b5a4`, standing **unverified**. It explicitly does not pin a complete source tree for those earlier commands. Final source-linked receipts will be additive; no earlier boundary becomes a final full-suite pass. Raw dependency/build/host logs and the native full ledger stay private.

Focused checks pin their actual source boundaries. Native statement `PCC-20261106-cc59ba` records the public lead, producer/reader and example observations, plus the separate standalone export boundary. Its standing is **unverified**; the root-library outcome and later reader-overflow hardening have their own continuation receipts [pcc:unverified]. Native statement `PCC-20261106-34db73` pins completed component checks, including the seven-test boundary before the later native-session test. Registry and optimization-policy repairs have their own subsequent validation; no earlier pass is transferred to them [pcc:unverified].

The subsequent final component receipt, native statement `PCC-20261106-5e324a`, records eight public lead tests, the existing producer bridges and examples, the repaired registry and optimization inventory, focused units and strict Rust 1.98/stable lints. Its standing is **unverified**. The native scratch session is a correctness check; APS receives a static audit and the rho control alone receives count replay. The broad library command remains a separate, earlier source boundary with failures and no final verdict in this receipt [pcc:unverified]. See CHECKS.md for the exact boundaries and limitations.

The algorithm/reader source-only pin is `413a93d6f9940c8552004681751741261366bb49c`. See the lead cells, trace pool, public producer bridge and inclusive reader. These links pin source, not an execution or performance verdict.

HANDOFF.md tracks the unfinished work. T9/research and T10/authblocked checkpoint stay open. Passing development tests or a merged implementation PR does not discharge the baseline/scaling request.

Sources checked: Amadori–Pintore–Sala, revised manuscript, sections 3.1–3.3; the previous source map, existing F4 record/replay, existing unequal-profile sampler and existing native Callgrind parser. KB unavailable: no loaded KB tool or local indexed runtime/catalog; the fallback is these primary and repository sources. No novelty claim.

Canonical scoreboard/curve/performance diagrams have no new measured value to add; they remain unchanged. This study supplies its own implementation and admission diagram and source-linked report, without a performance plot.

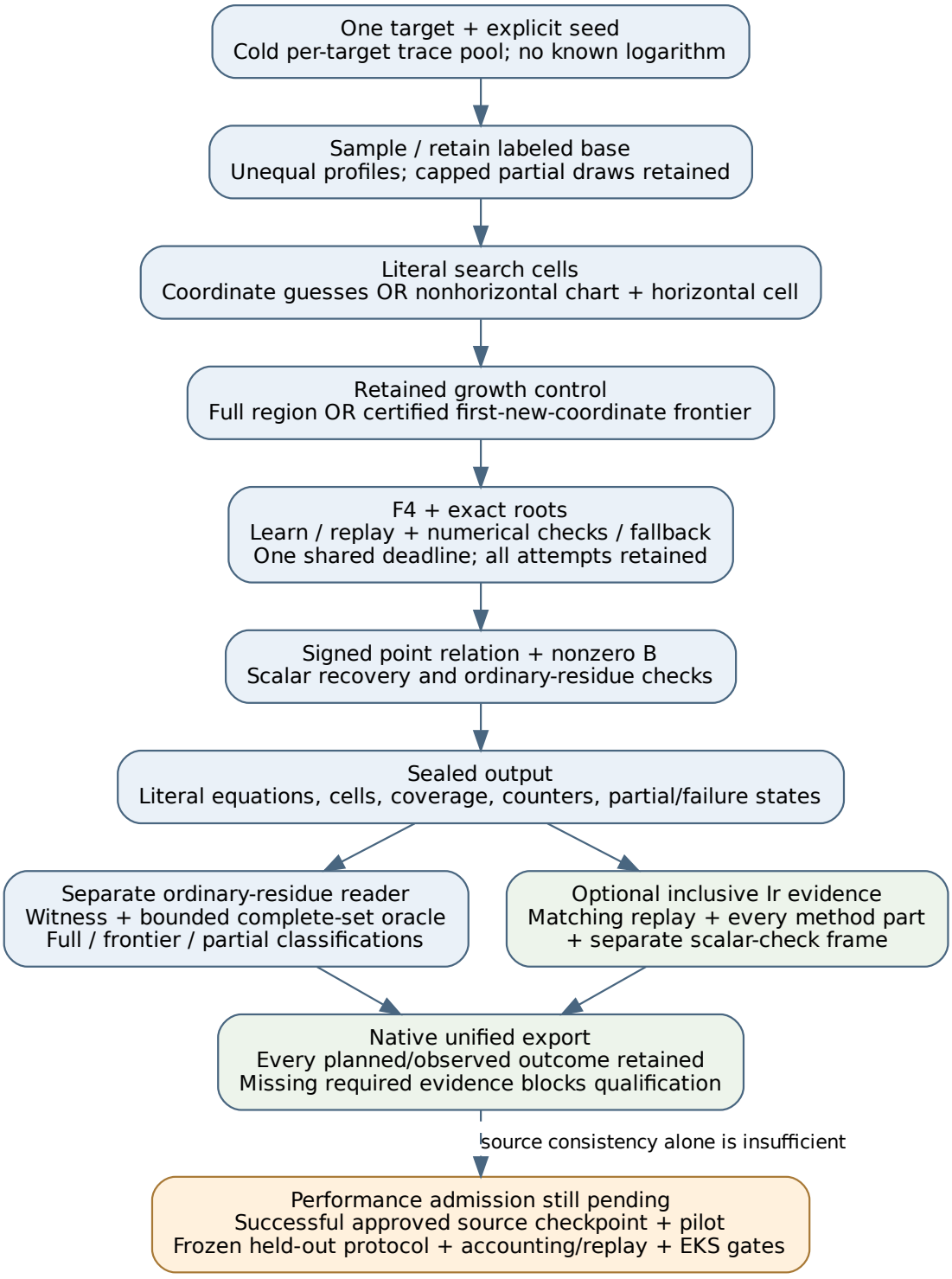


Figure 1: Implemented search paths and pending measurement admission